

Maths for ML: gradients, rank, contours

Gradients of linear and quadratic forms, rank and inverses, Jacobians, and contour plots.

THE QUESTION

How do we differentiate a loss with respect to a whole parameter vector, and what does the result look like?

THE INTUITION

Write the scalar out in components, take one partial per parameter and stack them in a column. On a contour plot that column is an arrow crossing the level lines at right angles, pointing uphill.

KEY TAKEAWAY

$\nabla a^T \theta = a$ and $\nabla \theta^T A \theta = (A + A^T) \theta$, which is $2A\theta$ for symmetric A . A square matrix is invertible exactly when its rank is full.

scalar loss $f(\theta)$ → one partial per θ_i → stack: gradient column → perpendicular to contours → step along -gradient

1 A sum of squares is a row times a column

$\epsilon^T \epsilon$ ϵ^T $1 \times N$ ϵ $N \times 1$

1	1	-2	2
-2			
2			

$\epsilon^T \epsilon = 1 + 4 + 4 = 9$
a 1×1 scalar: the sum of squared entries

$(AB)^T = B^T A^T$ the shapes agree
 A 2×3 · B 3×4 = AB 2×4
 $(AB)^T$ 4×2 = B^T 4×3 · A^T 3×2
 a scalar is its own transpose, so $a^T a = \theta^T a$

Vectors are columns, so $\epsilon^T \epsilon = \sum_i \epsilon_i^2$ for a residual vector ϵ . Transposes reverse a product, and a scalar is its own transpose. The derivative of a scalar s with respect to $\theta \in \mathbb{R}^N$ is the $N \times 1$ column of partials $\partial s / \partial \theta_i$: in ML the gradient has the shape of θ .

2 The gradient of $a^T \theta$ is a

$f(\theta) = a^T \theta = 2\theta_1 + \theta_2$
 $a = (2, 1)$, illustrative
 level sets: parallel lines
 $2\theta_1 + \theta_2 = c$, $c = -6$ to 6
 $\text{grad} = (2, 1) = a$ at every θ
 one arrow, at right angles to the lines

$a^T \theta = a_1 \theta_1 + a_2 \theta_2$ has partials a_1 and a_2 , so $\partial(a^T \theta) / \partial \theta = a$. The form is everywhere: $w^T x$ in linear regression, $w^T h + b$ in a neural network layer.

3 The gradient of $\theta^T A \theta$ is $(A + A^T) \theta$

$A = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 0 & 1 \end{bmatrix}$, illustrative
 $f(\theta) = \theta^T A \theta$, at $\theta = (1, 1)$
 $(A + A^T) \theta = (4, 8)$
 = finite differences; at right angles to the bold level set $f = 6$
 $2A\theta = (6, 6)$
 wrong: 71.6° to the contour

With $Z = \begin{pmatrix} e & f \\ f & g \end{pmatrix}$, $\theta^T Z \theta = e\theta_1^2 + 2f\theta_1\theta_2 + g\theta_2^2$, whose partials stack to $2Z\theta$. In general

$$\frac{\partial}{\partial \theta} \theta^T A \theta = (A + A^T) \theta,$$

which is $2A\theta$ only for symmetric A , such as $X^T X$ in the linear regression sheet ($\theta^T \theta$ gives 2θ). Finite differences check any formula:

```
A = np.array([[1., 2.], [0., 3.]])
f = lambda t: t @ A @ t # theta^T A theta
t, h = np.array([1., 1.]), 1e-5
fd = [(f(t + h*e) - f(t - h*e)) / (2*h)
        for e in np.eye(2)] # central diff.
print(np.round(fd, 3)) # [4. 8.]
print((A + A.T) @ t) # [4. 8.]
print(2 * A @ t) # [6. 6.]
```

4 Rank counts the independent rows

A 3×3 X 2×4

r1	0	1	2	
r2	1	2	1	
r3	2	7	8	

$r3 = 3 \cdot r1 + 2 \cdot r2$
 rank 2 · det 0 · max 3

1	2	4	4
3	4	8	0

rank at most $\min(2, 4) = 2$;
 rows not multiples of each other: rank 2

Rank is the largest number of linearly independent rows; counting columns gives the same number. An $r \times c$ matrix has rank at most $\min(r, c)$. In A , row 3 = 3 · row 1 + 2 · row 2, so the rank is 2, below the maximum of 3. The wide X has two rows, neither a multiple of the other, so it reaches its maximum, 2. In a data matrix, a column that is a combination of others adds no rank.

5 An outer product has rank 1

uv^T $2 \times -1 \times 1$ uv^T with $u = (1, 2, 3)$, $v = (2, -1, 1)$
 illustrative

1	2	-1	1
2	4	-2	2
3	6	-3	3

$2u \quad -1u \quad 1u$
 every column is a multiple of u , every row a multiple of v^T
 singular values: 9.17, 0, 0
 one nonzero: rank 1

uv^T (3×1 times 1×3) has entries $u_i v_j$. Column j is $v_j u$ and row i is $u_i v^T$, so every row and column lies along one vector: rank 1 for any nonzero u, v . The one nonzero singular value is $\|u\| \|v\|$.

6 Only a full-rank square matrix has an inverse

$M = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ $M = \begin{bmatrix} 1 & 2 & 0.5 \\ 1 & 1 & 1 \end{bmatrix}$

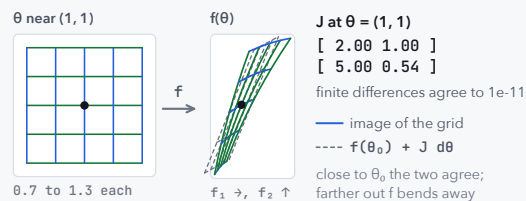
rank 2: the inverse $\begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & 1 \end{bmatrix}$ undoes it
 dashed: unit square; blue: its image
 rank 1: $(1, 0)$ and $(0, 0.5)$ both land on $(1, 0.5)$, so no inverse

A^{-1} satisfies $AA^{-1} = A^{-1}A = I_n$. An $n \times n$ matrix has one exactly when its rank is n , or equally $\det A \neq 0$; it is then nonsingular and the inverse is unique. A rank-deficient matrix squashes the plane onto a line, so two inputs share an output and nothing can undo it. The singular M here is the outer product $(1, 0.5)^T (1, 2)$; the A of panel 4 has $\det = 0$ too. The linear regression sheet inverts $X^T X$, which works only when the columns of X are independent; a repeated column makes it singular.

The gradient crosses every level set at a right angle

A vector output gives a Jacobian. Level sets turn f into a map, and the gradient points uphill across it.

7 The Jacobian: one gradient per output

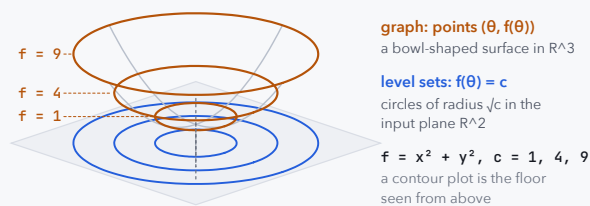


For $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, J is $m \times n$ with $J_{ij} = \partial f_i / \partial \theta_j$; row i is $(\nabla f_i)^T$. The lecture's example:

$$f = \begin{pmatrix} \theta_1^2 \theta_2 \\ 5\theta_1 + \sin \theta_2 \end{pmatrix}, \quad J = \begin{pmatrix} 2\theta_1 \theta_2 & \theta_1^2 \\ 5 & \cos \theta_2 \end{pmatrix}.$$

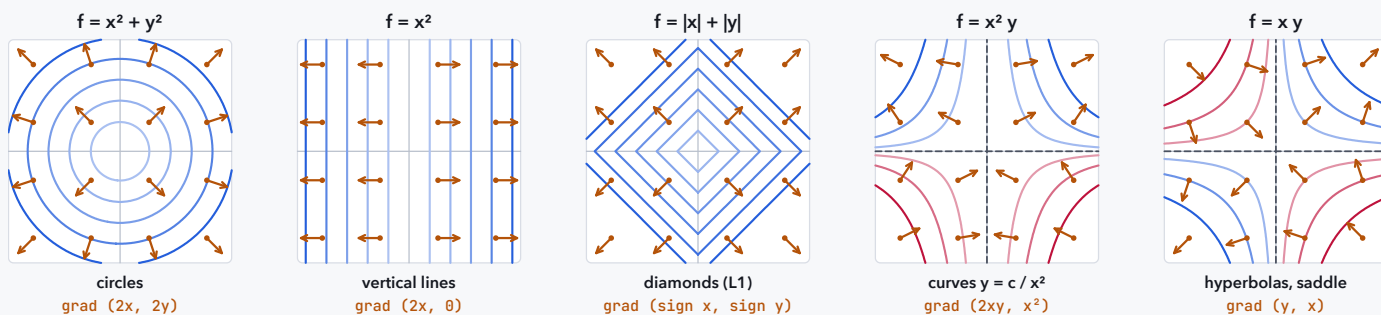
Near θ_0 , $f(\theta_0 + d\theta) \approx f(\theta_0) + J d\theta$. For $m = 1$, J is the gradient as a row.

8 A graph lives one dimension up



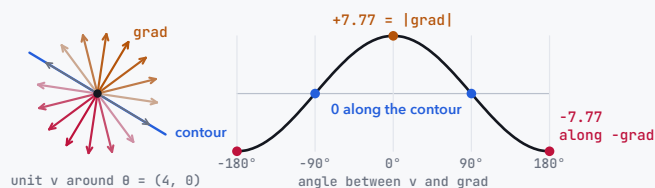
For $f: \mathbb{R}^n \rightarrow \mathbb{R}$ the graph $\{(\theta, f(\theta))\}$ lies in $\mathbb{R}^{n+1}$, a surface when $n = 2$. A level set $\{\theta: f(\theta) = c\}$ lies in the input space $\mathbb{R}^n$. To draw a contour plot, evaluate f on a grid (`np.meshgrid`) and trace where it crosses each c (here: `matplotlib`).

9 Five functions from the lecture, five contour shapes



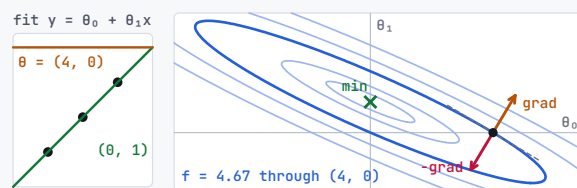
blue: positive levels, darker = higher · rose: negative levels · dashed: $f = 0$ · arrows: gradient direction (unit length), on a 4×4 grid of points x^2 ignores y , like a loss that ignores one parameter. $|x| + |y|$ (L1, behind lasso) has no gradient on the axes, at the diamonds' corners. $x^2 y$ has zero gradient on the whole y -axis. xy has a saddle at the origin: zero gradient, yet f rises along $y = x$ and falls along $y = -x$. Saddles are common in neural network losses.

10 Why the gradient meets contours at 90°



On a curve $c(t)$ inside a level set, $f(c(t)) = K$, so by the chain rule $0 = \nabla f \cdot c'(t)$. For a unit direction v , $D_v f = \nabla f \cdot v = \|\nabla f\| \cos \varphi$: largest along ∇f (steepest ascent), zero along the contour, most negative along $-\nabla f$.

11 A line-fit loss has elliptical contours



The contour notebook's mean squared error for the three points is a quadratic form, $f = \theta^T A \theta + b^T \theta + 14/3$ with $A = \begin{pmatrix} 1 & 2 \\ 2 & 14/3 \end{pmatrix}$ and $b = (-4, -28/3)$. So $\nabla f = 2A\theta + b = (4, 6.67)$ at $(4, 0)$, the notebook's autograd value.

12 Pause and think

Gradient of $\theta_1 \theta_2 = \theta^T \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \theta$
 (θ_2, θ_1) ; $2A\theta = (2\theta_2, 0)$ is wrong.
 Nearest point to 0 on $x + y = 1$?
 $(0.5, 0.5)$, where a circle touches the line; distance 0.707 .

Can a 3×5 matrix have rank 4? No: rank is at most $\min(3, 5) = 3$.

Is a zero gradient a minimum? Not always: xy has a saddle there.

Gradient of $|x| + |y|$ at $(1, -2)$?
 $(1, -1)$, straight out of the diamond.

Which way does gradient descent step? Along $-\nabla f$, across the contours.

$\nabla a^T \theta = a \cdot \nabla \theta^T A \theta = (A + A^T) \theta$ · invertible when rank is full · the gradient is perpendicular to the level set

Next: the linear regression and gradient descent sheets. Source: lecture slides "Maths for ML" and "Contour plots and gradients", and the contour and maths-for-ml-2 notebooks (nipunbatra.github.io/ml-teaching). Vectors and matrices marked illustrative are ours.